

试卷分析 (以 A 卷为例)

上 海 交 通 大 学 试 卷 (A 卷)

(2024 至 2025 学 年 第 二 学 期)

班级号 _____ 学号 _____ 姓名 _____

课程名称 _____ 高等代数 (荣誉) II _____ 成绩 _____

第一题 (A 卷题目)

- (1) Let $U \subseteq V$ be a subspace. Let X be a set of all subspaces of the quotient space V/U and Y be a set of all subspaces of V that contains U . Then there is a one-to-one correspondence between X and Y .

正确. 这是同构定理 C 的表述, 见[作业原题](#).

- (2) Let U, V, W be three finite dimensional vector spaces over $\mathbb{F}$, and $\varphi : V \rightarrow W$ be a linear transformation. If φ is injective, then there is an induced natural linear transformation $\text{Hom}(U, V) \rightarrow \text{Hom}(U, W)$ which is also injective.

正确. f 单时 $f \circ -$ 单. 见[作业原题](#).

- (3) Let V be a finite dimensional vector space over a field $\mathbb{F}$ and $W \subseteq V$ be a subspace. Let $\langle, \rangle$ be a nonsingular bilinear form and $W^\perp := \{v \in V, s.t. \langle v, w \rangle = 0 \text{ for all } w \in W\}$. Then one always has $W \oplus W^\perp = V$.

错误. 反例就在试卷上: 见 ω 所在的题目.

- (4) Let V be a Euclidean space and $\mathbf{v}_1, \dots, \mathbf{v}_m$ be a set vectors in V . Denote by $g_{ij} = \langle \mathbf{v}_i, \mathbf{v}_j \rangle$ the inner product of $\mathbf{v}_i$ and $\mathbf{v}_j$. Consider the Gram matrix of $\mathbf{v}_1, \dots, \mathbf{v}_m$, which we denote by $G = (g_{ij})_{i,j=1,\dots,m}$. Then the Gram matrix G is always positive definite.

错误. 取 $\mathbf{v}_i$ 全零.

第一题 (B 卷题目)

- (1) Let V be a finite dimensional vector space over a field $\mathbb{F}$ and $W \subseteq V$ be a subspace. Let $\langle, \rangle$ be a nonsingular bilinear form and $W^\perp := \{v \in V, s.t. \langle v, w \rangle = 0 \text{ for all } w \in W\}$. Then one always has $W \oplus W^\perp = V$.

错误. 反例就在试卷上: 见 ω 所在的题目.

- (2) Let $U \subseteq V$ be a subspace. Let X be a set of all subspaces of the quotient space V/U and Y be a set of all subspaces of V that contains U . Then there is a one-to-one correspondence between X and Y .

正确. 这是同构定理 C 的表述, 见[作业原题](#).

- (3) Let V be a n -dimensional vector space over the field $\mathbb{F}$ and $\{\alpha_1, \dots, \alpha_n\}$ be a basis of V . Consider a bilinear form $\langle, \rangle$ on V , and let $G = (\langle \alpha_i, \alpha_j \rangle)_{i,j=1}^n$. For any vector $\mathbf{x}$, define $\mathbf{x}^\perp = \{y \in V \mid \langle \mathbf{x}, y \rangle = 0\}$. Then the matrix is invertible if and only if $\mathbf{x}^\perp \subsetneq V$ is a real subspace of V for $\mathbf{x} \neq 0$.

正确. 使用 $x \in \mathbb{F}^n$ 中向量表示 $\mathbf{x} \in V$ 在基下的坐标, 则 $\langle \mathbf{x}, \mathbf{y} \rangle = x^T G y$. 此时, G 可逆当且仅当 $\text{null}(G) = 0$, 即 $\mathbf{x}^\perp$ 总是真子空间 ($\mathbf{x} \neq \mathbf{0}$).

- (4) Let U, V, W be finite dimensional vector spaces over $\mathbb{F}$ and $\varphi : V \rightarrow W$ be a linear transformation. If φ is surjective, then there is an induced natural **surjective** linear map $\text{Hom}(W, U) \rightarrow \text{Hom}(V, U)$.

错误. f 满时 $- \circ f$ 单. 见[作业原题](#).

第二题

二、(15 points) Let V be a **finite** dimensional vector space over a field $\mathbb{F}$, and $W \subseteq V$ a subspace. Denote by V^* the dual space of V , and $W^\circ = \{f \in V^*, \text{ s.t. } f(w) = 0, \forall w \in W\} \subseteq V^*$ the annihilator of W . Please prove the following statements.

(1) (5 points) There is a natural isomorphism $(V/W)^* \rightarrow W^\circ$.

回答 I. 说明这是商空间的泛性质: 映射 $f \in V^*$ 能下降至商空间 V/W , 当且仅当 $f(W) = 0$.

回答 II. 使用 **coker**-对象的定义: $\text{Hom}(\text{coker}(i), \mathbb{F}) \simeq \ker \text{Hom}(i, \mathbb{F})$, 此处 $i: W \subseteq V$. 见[作业原题](#).

回答 III. 见恒等式 L2 的证明, 见[作业原题](#).

(2) (5 points) There is a natural isomorphism $V^*/W^\circ \rightarrow W^*$.

即恒等式 L1 的证明, 见[作业原题](#). 由于无限维情况需要 HB 公理, 故此体**必须使用维数或基扩张论证**.

(3) (5 points) Let W_1, W_2 be two subspaces of V such that $W_1 \oplus W_2 = V$. Then one has $W_1^* \cong W_2^\circ$ and $W_2^* \cong W_1^\circ$.

该问没有对自然性的要求. 有限维空间的同构等价于维数相同.

第三题

三: (15 points) Let V be an m dimensional vector space over $\mathbb{R}$. Suppose that there is a nonsingular skew-symmetric bilinear form on V , which is denoted by $\omega(u, v) : V \times V \rightarrow \mathbb{R}$. Such a vector space together with the bilinear form ω is called a symplectic vector space. Please prove the following facts about symplectic vector spaces.

这是去年强基班的作业原题 (反对称空间之 Schmidt-正交化类似物), 参阅 [2023-2024 春 作业题集, Problem 69](#).

- (1) (5 points) Prove that there exists a pair of vectors $\alpha, \beta \in V$, such that $\omega(\alpha, \alpha) = 0, \omega(\beta, \beta) = 0, \omega(\alpha, \beta) = 1$.

任取 $\alpha \neq 0$. 由 ω 非退化, 得 $\omega(\alpha, -) \neq 0$, 即存在 β 使得 $\omega(\alpha, \beta) = 1$. 反对称自动保证 $\omega(x, x) = 0$.

- (2) (5 points) Prove that V admits a basis $\{\alpha_1, \beta_1, \alpha_2, \beta_2, \dots, \alpha_n, \beta_n\}$ such that $\omega(\alpha_i, \alpha_j) = 0, \omega(\beta_i, \beta_j) = 0, \omega(\alpha_i, \beta_j) = \delta_{i,j}$ for $i, j = 1, \dots, n$. Further show that the dimension m of V must be even.

记 $(\alpha_1, \beta_1) := (\alpha, \beta)$. 下证

$$V = \underbrace{[\ker \omega(\alpha_1, -) \cap \ker \omega(\beta_1, -)]}_{\text{记作 } V_1} \oplus \underbrace{\text{span}(\alpha_1, \beta_1)}_{\text{记作 } W_1}.$$

易检验 $V_1 \cap W_1 = 0, \dim V_1 \geq (m - 2)$, 以及 $\dim W_1 = 2$. 遂得直和关系.

由构造, $\omega(w, v) = 0$ 对一切 $(v, w) \in V_1 \times W_1$ 成立. 对新得的双线性空间 (V_1, ω) 使用题 (1), 并记 $(\alpha_2, \beta_2) := (\alpha, \beta)$.

归纳得证.

- (3) (5 points) A matrix $\Omega \in \mathcal{L}(V)$ is called symplectic, if it preserves the symplectic structure, which means $\omega(\Omega\alpha, \Omega\beta) = \omega(\alpha, \beta)$ for all $\alpha, \beta \in V$. Prove that the determinant of a symplectic matrix is always 1.

回答 I. 使用数学分析的外积. 记 $\{d_i, e_i\}_{i=1}^n$ 是 V 的对偶基, 式 $\omega(-, ?) = \omega(\Omega(-), \Omega(?))$ 表明

$$\sum_{i=1}^n (d_i \circ \Omega) \wedge (e_i \circ \Omega) = \sum_{i=1}^n d_i \wedge e_i.$$

对上式两侧作 n 次外积 ($\eta \mapsto \underbrace{\eta \wedge \cdots \wedge \eta}_{n \uparrow \eta}$), 得

$$\begin{aligned} & (d_1 \circ \Omega) \wedge \cdots \wedge (d_n \circ \Omega) \wedge (e_1 \circ \Omega) \wedge \cdots \wedge (e_n \circ \Omega) \\ &= d_1 \wedge \cdots \wedge d_n \wedge e_1 \wedge \cdots \wedge e_n. \end{aligned}$$

左式是右式的 $\det \Omega$ 倍, 故 $\det \Omega = 1$. 可参考[同济教材](#)中的涉及行列式与体积形式的章节.

回答 II. 以上标准基对应的 Gram 矩阵是 $J := \begin{pmatrix} O & I \\ -I & O \end{pmatrix}$. 式 $\omega(-, ?) = \omega(\Omega(-), \Omega(?))$ 即 $\Omega^T J \Omega = J$. 上学期[作业原题](#)给出了简单证明, 以及更精细的岩澤(いわさわ)分解.

回答 III. 依照反对称矩阵的 Pfaffian, 得 $\det \Omega \cdot \text{Pf}(J) = \text{Pf}(J) \neq 0$, 即 $\det \Omega = 1$.

第四题

四:(20 points) Let U, V be two vector spaces over a field $\mathbb{F}$, and $W \subset V$ a subspace. No assumptions on dimensions.

- (1) (10 points) Let $A = \{T \in \mathcal{L}(V, U) \mid W \subseteq \ker(T)\} \subseteq \mathcal{L}(V, U)$ be a subspace in the space of linear transformations from V to U . Prove that $A \cong \mathcal{L}(V/W, U)$ and $\mathcal{L}(V, U)/A \cong \mathcal{L}(W, U)$ by constructing explicit isomorphisms.

见[作业原题](#). 同构式 $\mathcal{L}(V, U)/A \cong \mathcal{L}(W, U)$ 的说明需要额外公理, 以下是两种可以得满分的回答:

1. 硬气地表明自己不承认选择公理, 指出此题就是错题.
2. 同[作业题解答](#), 道明**关键系额外公理** (例如选择公理, 或是弱一些的 (co)flasque 条件).

- (2) (10 points) Let $B = \{T \in \mathcal{L}(U, V) \mid \text{im}(T) \subseteq W\} \subseteq \mathcal{L}(U, V)$ be a subspace in the space of linear transformations from U to V . Prove that $B \cong \mathcal{L}(U, W)$, and $\mathcal{L}(U, V)/B \cong \mathcal{L}(U, V/W)$.

原题所在位置与作答方式同上.

第五题

五: (20 points) Let U, V be two vector spaces over a field $\mathbb{F}$ which are not necessarily finite-dimensional.

(1) (5 points) Please construct a natural linear map $\alpha : U^* \otimes_{\mathbb{F}} V \rightarrow \text{Hom}(U, V)$.

双线性 $B : U^* \times V \rightarrow \text{Hom}(U, V)$, $(f, v) \mapsto [u \mapsto f(u) \cdot v]$ 诱导了 α , 使得简单张量 $f \otimes v$ 之像是 $[u \mapsto f(u) \cdot v]$.

(2) (5 points) Please prove that the image space of the above linear map is the subspace $\{\phi \in \text{Hom}(U, V) \mid \dim(\text{im}(\phi)) < \infty\}$.

本题与张量无关, 因为 α 之像空间即双线性 B 之像的张成空间, 亦即全体秩 1 映射之张成空间.

(3) (5 points) Please prove that the linear map α is always injective.

任取 $f := \sum_{i=1}^d f_i \otimes v_i$, 不妨设 $\{f_i\}$ 与 $\{v_i\}$ 均是线性无关集.

假定 $\alpha(f) = 0$, 往证 $f = 0$. 由定义, $\alpha(f)$ 给出一族零线性组合式

$$\left\{ \sum_{i=1}^d f_i(u) \otimes v_i = 0 \right\}_{u \in U}.$$

由 $\{v_i\}$ 线性无关, 得一族零向量 $\{f_i(u) = 0\}_{u \in U, 1 \leq i \leq d}$. 因此诸 $f_i = 0$.

(4) (5 points) Please prove that when both U and V are finite dimensional, the linear map α is an isomorphism.

将 α 分解作 $U^* \otimes V \xrightarrow{p} \text{im}(\alpha) \xrightarrow{i} \text{Hom}(U, V)$, 既证 p 为同构. 若 $\dim U < \infty$ 或 $\dim V < \infty$, 则 i 满, 即 α 是同构.

第六题

六、(10 points) Let U, V, W, Z be vector spaces over $\mathbb{F}$. Assume that we have a sequence of homomorphisms

$$U \xrightarrow{g} V \xrightarrow{f} W \rightarrow 0$$

such that f is surjective and $\text{im}(g) = \ker(f)$. We call such a sequence exact.

Consider the tensor product of the above sequence with Z :

$$U \otimes Z \xrightarrow{g \otimes \text{Id}_Z} V \otimes Z \xrightarrow{f \otimes \text{Id}_Z} W \otimes Z \rightarrow 0.$$

Prove that the new sequence is exact, which means:

(1) (5 points) $f \otimes \text{Id}_Z$ is surjective,

$W \otimes Z$ 中任意张量 $\sum_{\text{有限和}} w_i \otimes z_i$ 总有原像 $\sum_{\text{有限和}} v_i \otimes z_i$, 其中 $f(v_i) = w_i$.

(2) (5 points) $\text{im}(g \otimes \text{Id}_Z) = \ker(f \otimes \text{Id}_Z)$.

由 coker 对象之定义 (见[作业原题](#)), 仅需说明对一切线性空间 X , 总有 $\ker$ 对象

$$\underbrace{\text{Hom}(W \otimes Z, X)}_{=\ker(- \circ (g \otimes 1_Z))?} \xrightarrow[\text{单}]{- \circ (f \otimes 1_Z)} \text{Hom}(V \otimes Z, X) \xrightarrow{- \circ (g \otimes 1_Z)} \text{Hom}(U \otimes Z, X).$$

以自然同构

$$\text{Hom}(- \otimes !, @) \simeq \text{Hom}(-, \text{Hom}(!, @)) \xlongequal{\text{简化地记作}} (-, (!, @))$$

等价转换上式, 即证

$$\underbrace{(W, (Z, X))}_{=\ker(- \circ g)?} \xrightarrow{- \circ f} (V, (Z, X)) \xrightarrow{- \circ g} (U, (Z, X)).$$

将 (Z, X) 视作整体, 上式无非 $\text{coker}(g) \simeq W$ 之推论.